

WPI

LLM Profiling and Fine-Tuning with Limited Neighbor Information for Node Classification on Text-Attributed Graphs

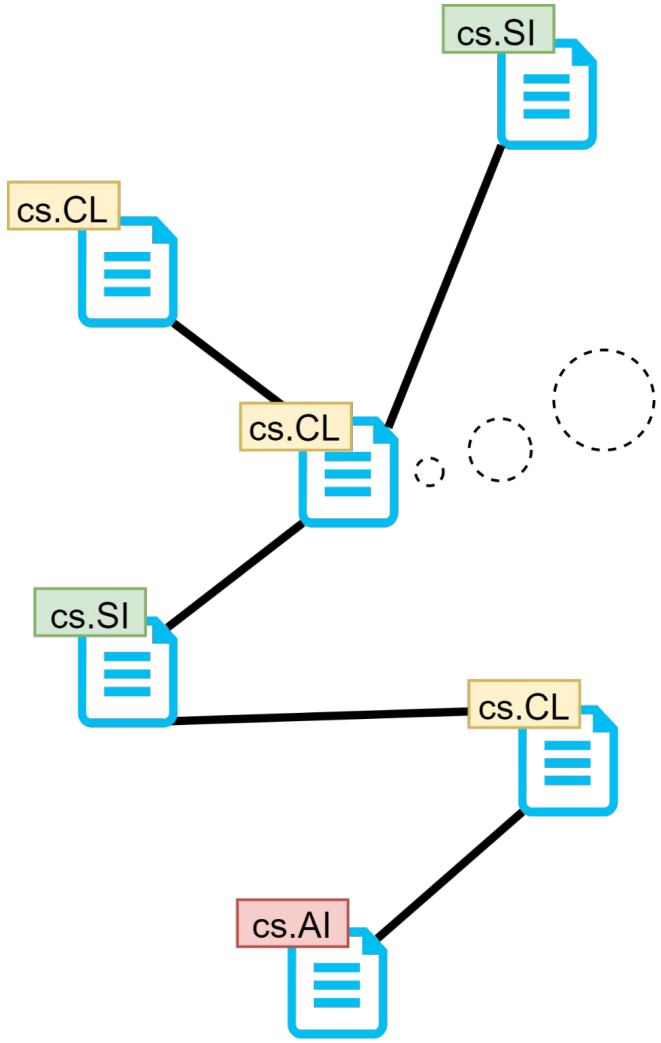
Xinyi Fang, Kyumin Lee, Yichuan Li



Background and Related Work

Text-Attributed Graphs
Task Description: Node Classification
Example of Text Data Augmentation
Related Work

Text-Attributed Graphs



LLM Profiling and Fine-Tuning with Limited Neighbor Information for ...

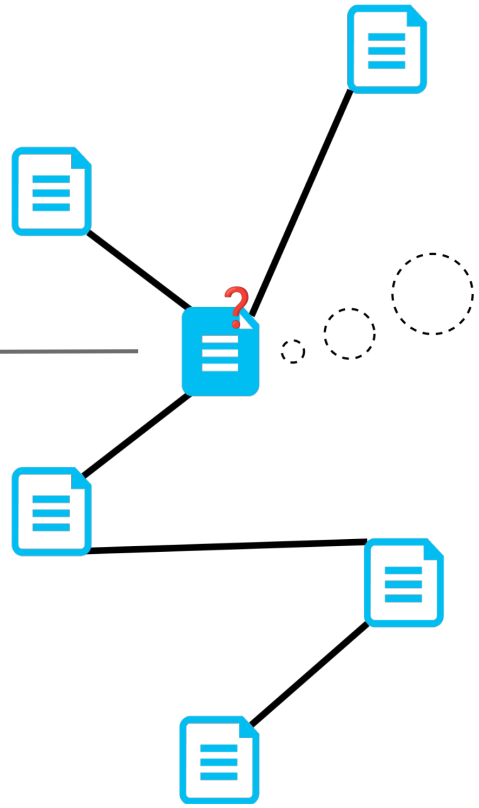
Graph representation learning has become a critical task across various domains such as social networks and recommender systems. Recently, the rise of large language models (LLMs) has opened up new possibilities for processing text-attributed graphs (TAGs), where nodes are associated with textual information. Despite promising progress, applying LLMs to TAGs faces significant challenges, including input window size limitations and the computational overhead of handling large-scale graphs. To address these challenges, we propose a novel approach that leverages a multi-profiling framework as a data augmentation method, thereby increasing the diversity and quantity of the training samples. The profiles/summaries generated by each of the five profiling/summarizing models are then combined with the graph structure ...

Nodes contain unstructured text

Task: Node Classification

Classes
cs.CL
cs.SI
cs.AI
...

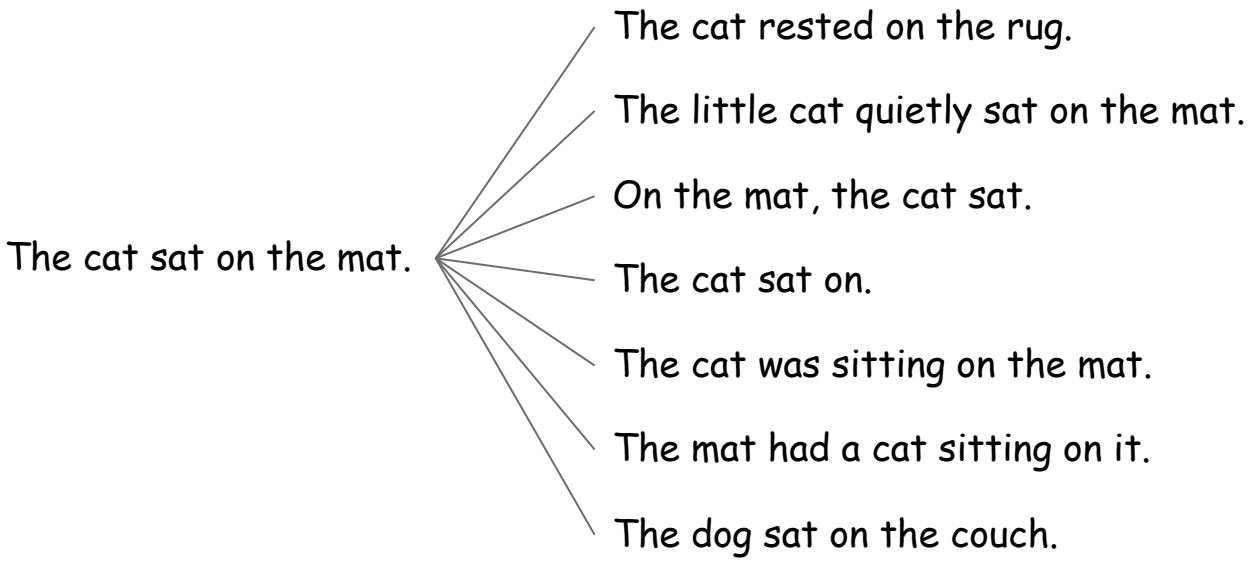
classify the node into the correct category



LLM Profiling and Fine-Tuning with Limited Neighbor Information for ...

Graph representation learning has become a critical task across various domains such as social networks and recommender systems. Recently, the rise of large language models (LLMs) has opened up new possibilities for processing text-attributed graphs (TAGs), where nodes are associated with textual information. Despite promising progress, applying LLMs to TAGs faces significant challenges, including input window size limitations and the computational overhead of handling large-scale graphs. To address these challenges, we propose a novel approach that leverages a multi-profiling framework as a data augmentation method, thereby increasing the diversity and quantity of the training samples. The profiles/summaries generated by each of the five profiling/summarizing models are then combined with the graph structure ...

Text Data Augmentation



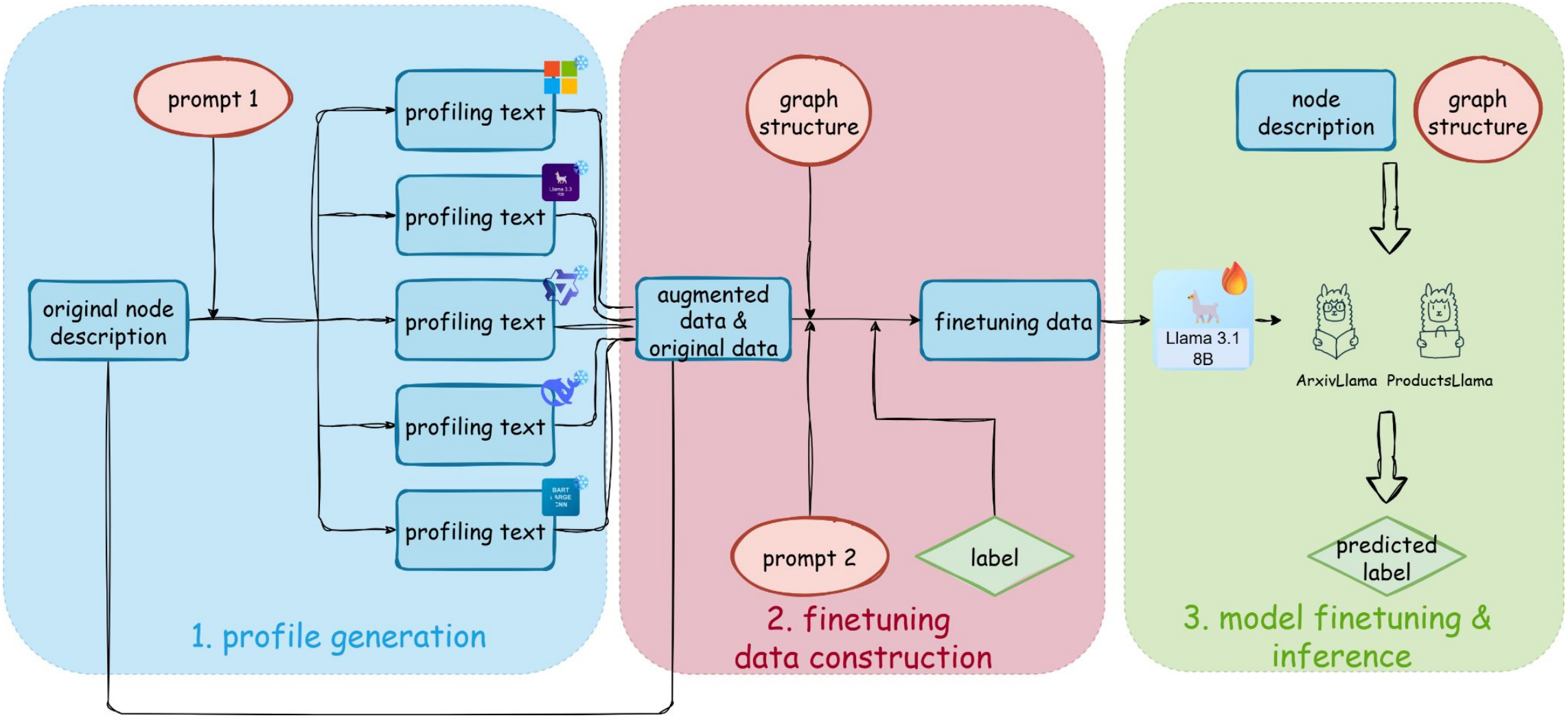
Related Work

- Methods for Text-attributed Graphs
  - GNN-based Methods
  - LLMs combine with GNNs
  - Transformer-based Methods
  - Limitations to current LLMs for text-attributed graphs: effective but is limited to small-scale datasets due to the context window limit
- Data Augmentation
  - LLM-based data augmentation: rewrite training samples[1], generate context and question-answer pairs[2], use LLM as a substitute for human annotators[3]
  - Data augmentation using different pre-trained models varies in terms of data diversity[4]
- Finetuning for LLMs
  - Techniques for accelerating finetuning: LoRA[5], QLoRA[6] and rsLoRA[7]
  - Tools and frameworks: Unsloth[8]

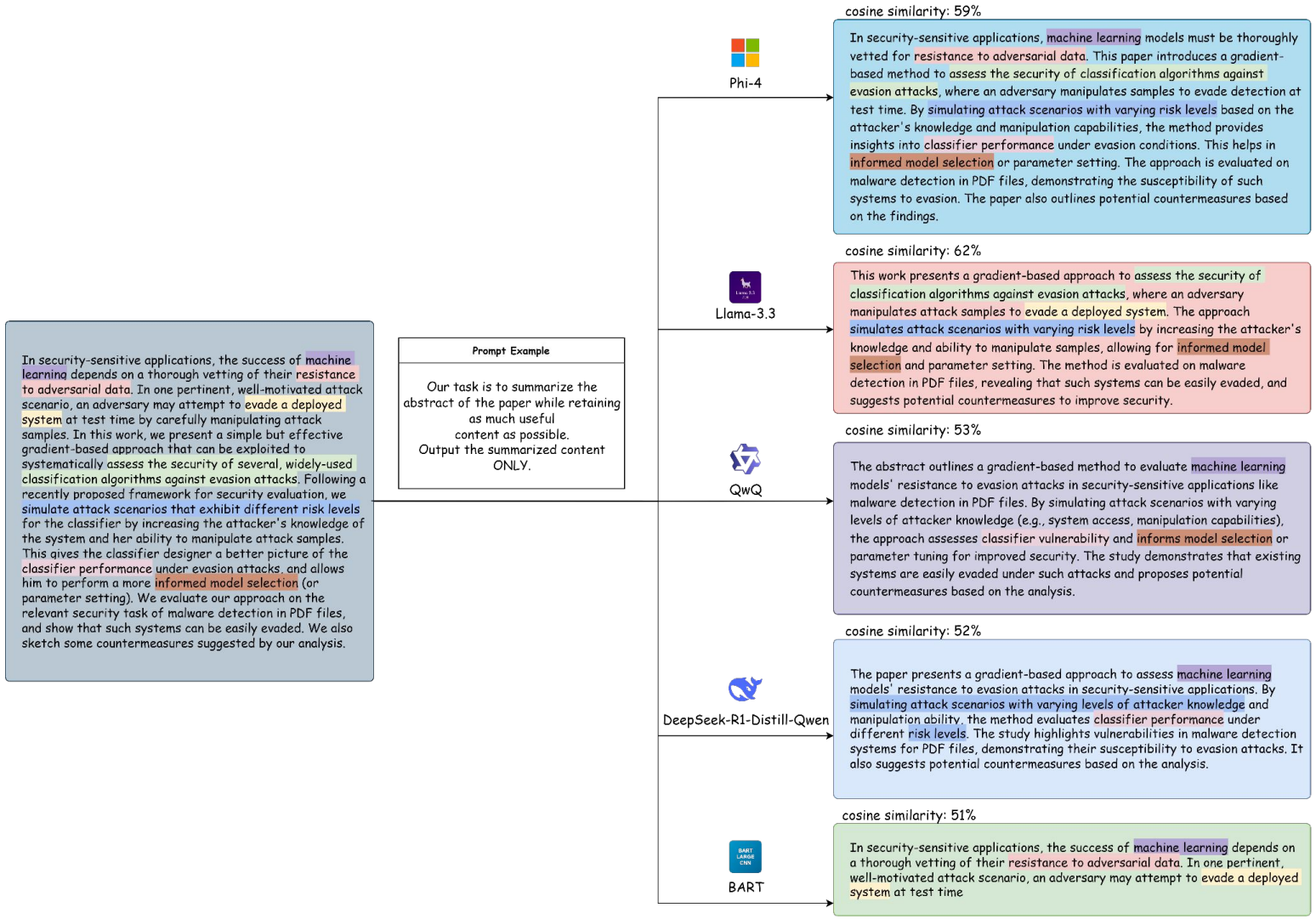
Our Method

The Overall Structure
Data Augmentation through LLMs' Profiling
Construction of Finetuning Data

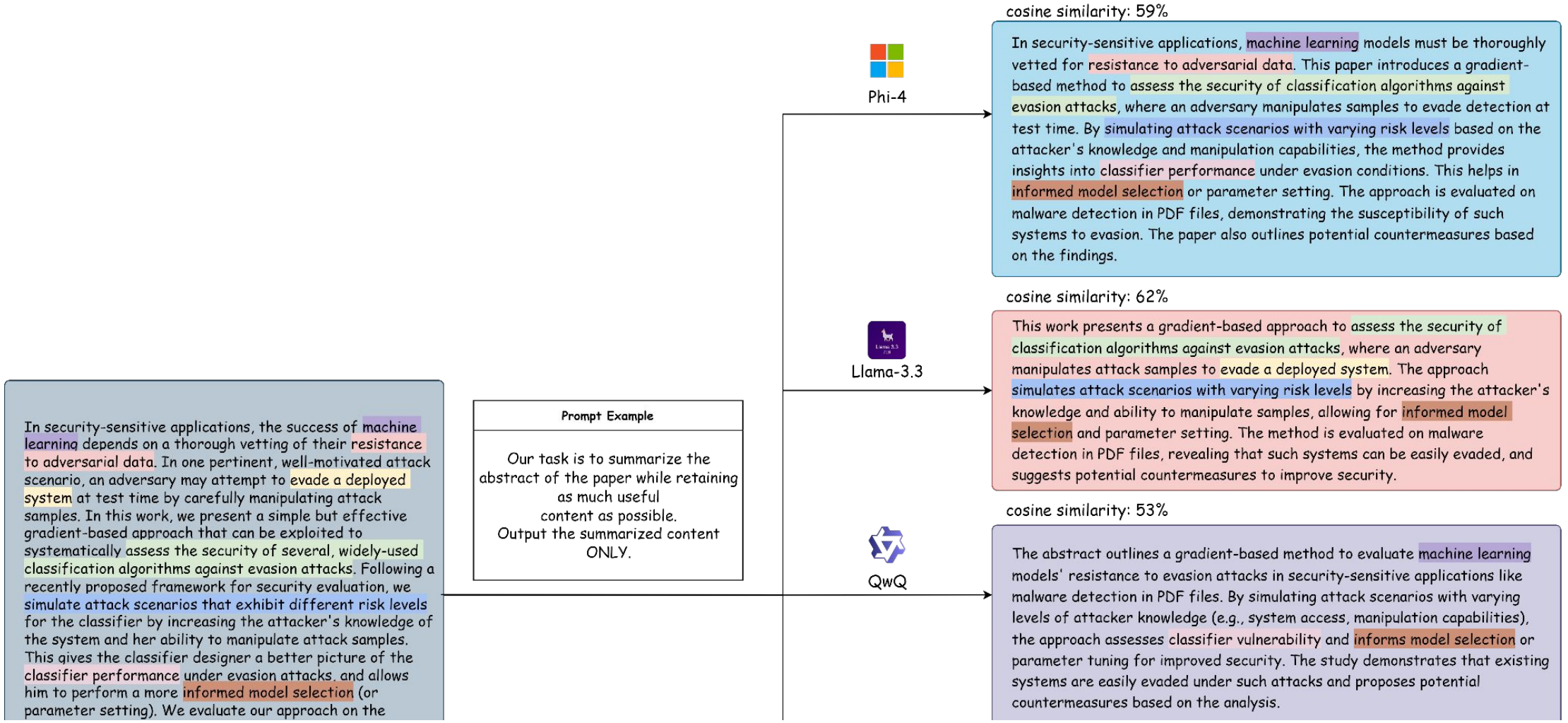
The Overall Structure



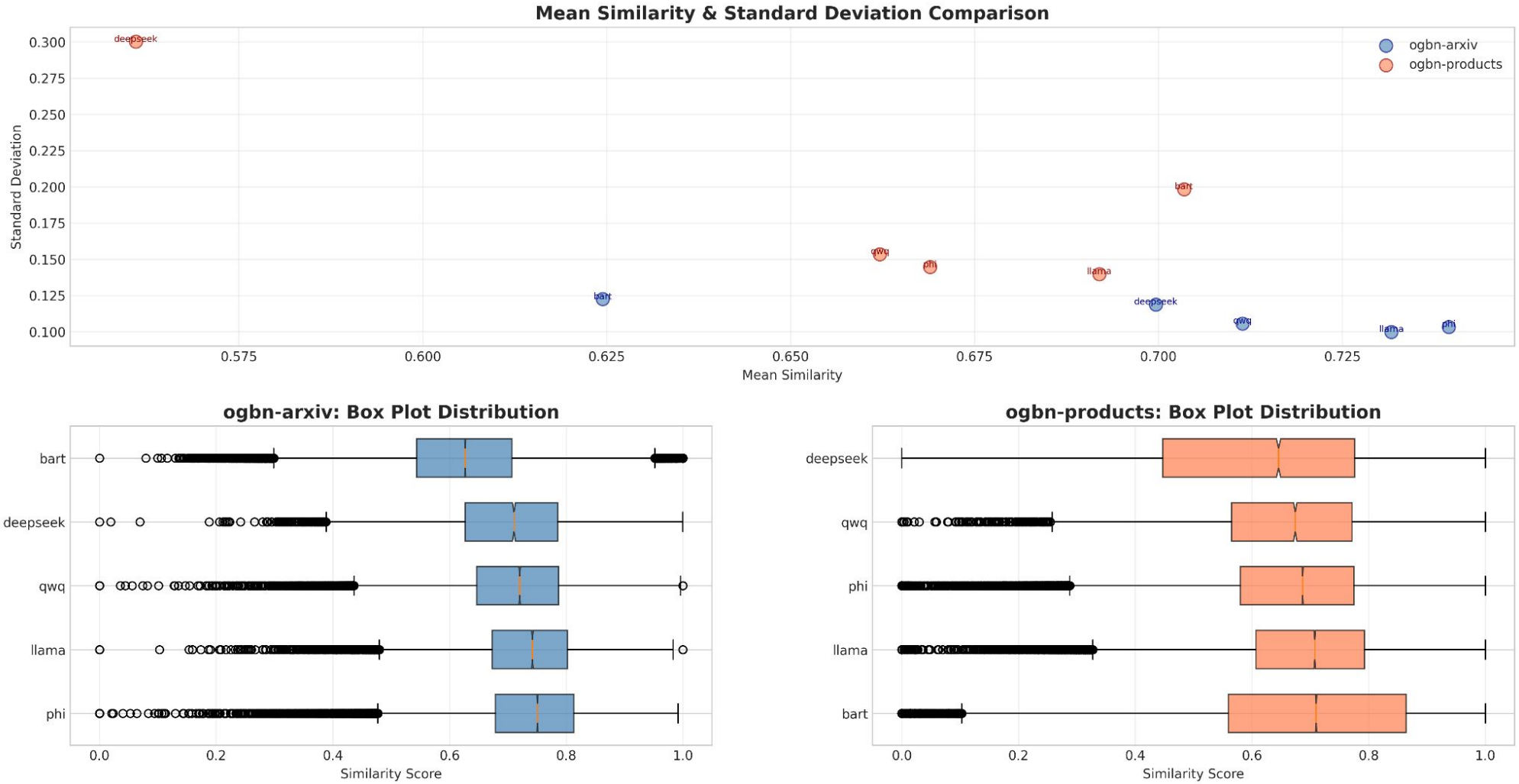
Data Augmentation through LLMs' Profiling



Data Augmentation through LLMs' Profiling



Data Augmentation through LLMs' Profiling



Construction of Finetuning Data

Arxiv Prompt Example for Finetuning

<|im_start|>system

You are a classifier that determines which category a given node belongs to from arxiv cs. The 40 categories are: 0-(Numerical Analysis) 1-(Multimedia) ... 38-(Digital Libraries) 39-(Discrete Mathematics). Output the corresponding number ONLY

<|im_end|>
<|im_start|>user

Node Title: T_v Abstract: A_v
Neighbors:
- Node Title: T_u (h hop) Abstract: A_u
...

<|im_end|>
<|im_start|>assistant

y_v

<|im_end|>

Algorithm 1 Node Description Construction for Arxiv Dataset

procedure DESCRIPTIONCONSTRUCTION(node v, max hops $h_{\max}$, max neighbors $n_{\max}$)
  Initialize $D[v] \leftarrow$ "Node Title: 'T_v'\nAbstract: 'A_v'\n\nNeighbors:\n"
  Initialize an empty set S to track visited nodes
  Initialize a queue Q with the node and its initial hop count $(v, 0)$
  Initialize neighbor_count $\leftarrow 0$
  while Q is not empty **and** neighbor_count $< n_{\max}$ **do**
    Remove the first element (u, h) from Q
    if u is already visited **or** $h > h_{\max}$ **then**
      continue
    end if
    Mark u as visited by adding it to S
    if $u \neq v$ **then**
      Append neighbor information to $D[v]$:
      $D[v] \leftarrow D[v] +$ "- Node Title: 'T_u' (h hop)\n Abstract: 'A_u'\n"
      Increment neighbor_count
    end if
    for all neighbors w of u **do**
      Add $(w, h+1)$ to the end of Q
    end for
  end while
end procedure

Experiments and Results

Dataset Description
The Amount of Profiling Data
Impact of Profiling Models
Influence of Neighbor Node Quantity
Training Parameters and Hardware Specifications

Dataset Description

Dataset	Nodes	Edges	Labels	Evaluation Metric	Split Ratio(%)
ogbn-arxiv	169,343	1,166,243	40	Accuracy	53.7/17.6/28.7
ogbn-products	2,449,029	61,859,140	47	Accuracy	8.03/1.60/90.37

- ogbn-arxiv
  - the task is to predict the 40 subject areas of arXiv CS papers, such as cs.AI, cs.LG, and cs.OS
  - split the data as follows:
    - train on papers published until 2017
    - validate on papers published in 2018
    - test on papers published since 2019
- ogbn-products
  - the task is to predict the category of a product from the 47 top-level categories
  - split data based on sales rank (popularity):
    - the top 8% of products by rank are used for training
    - the next 2% of products are used for validation
    - the remaining products are used for testing

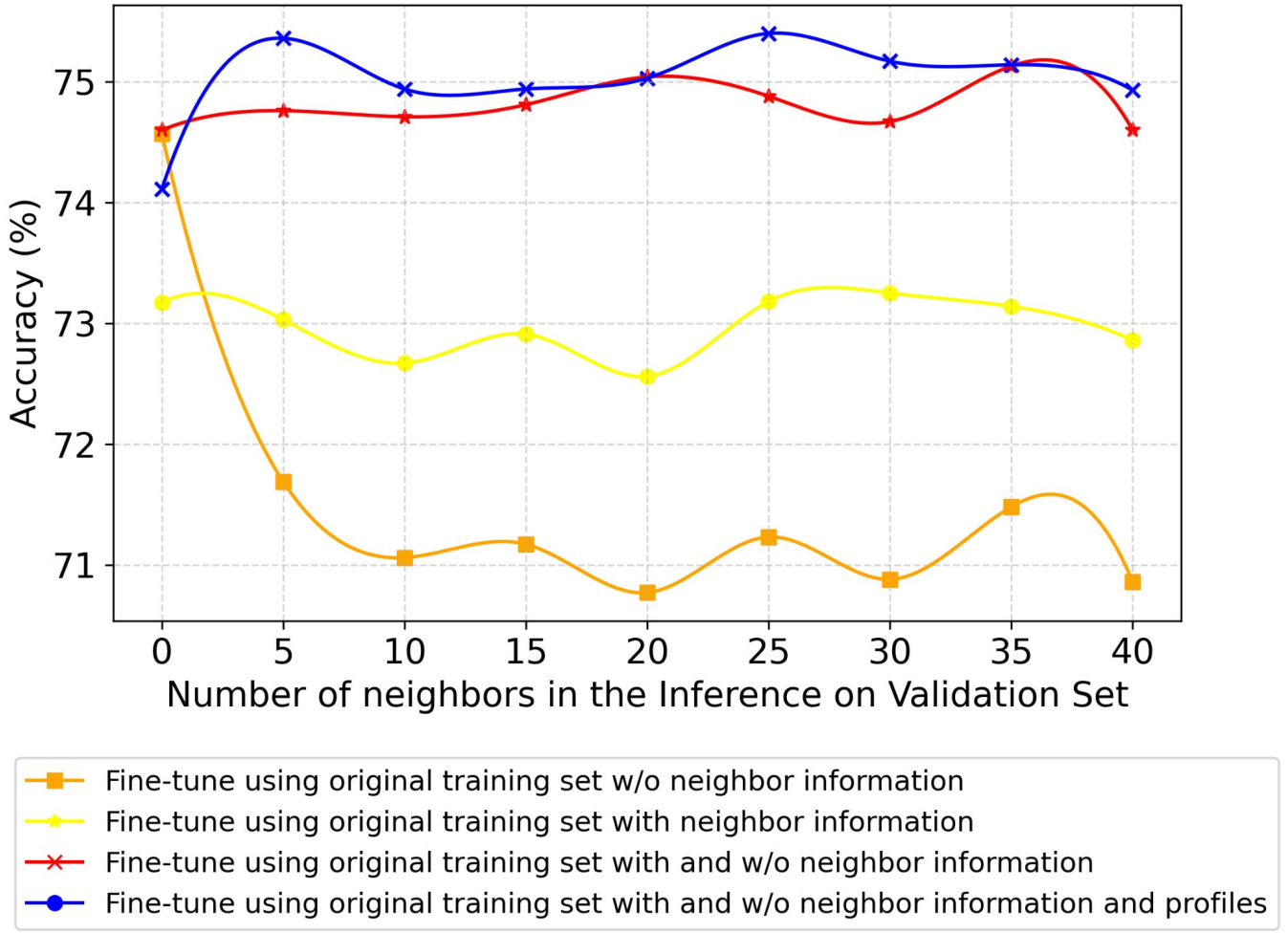
The Amount of Profiling Data

Dataset	Nodes	Edges	Labels	Evaluation Metric	Split Ratio(%)	Training Data
ogbn-arxiv	169,343	1,166,243	40	Accuracy	53.7/17.6/28.7	91k
ogbn-products	2,449,029	61,859,140	47	Accuracy	8.03/1.61/90.37	197k

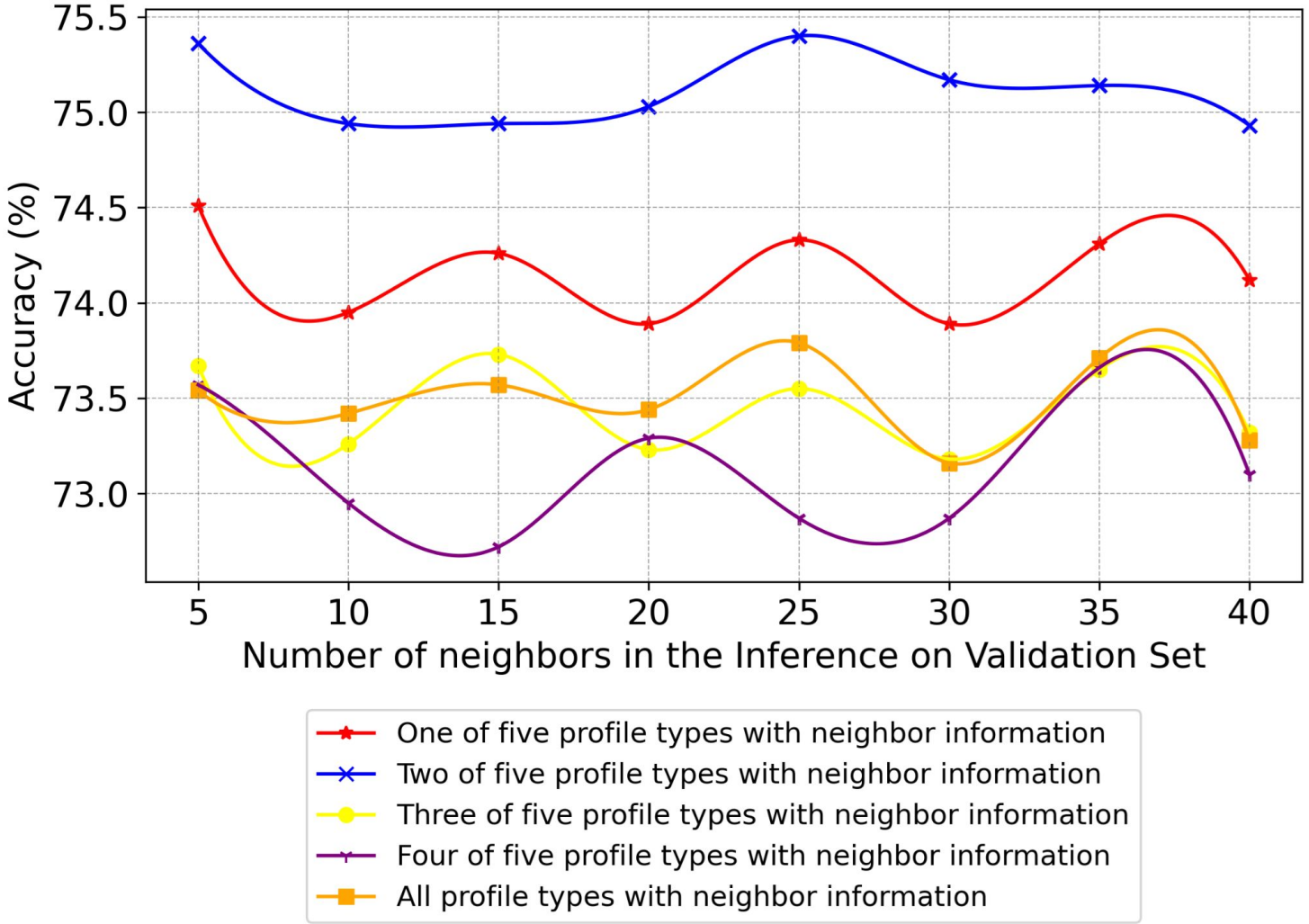
The amount of data profiled/augmented by each model:

	Phi-4	Llama-3.3-70b	QwQ	DeepSeek-R1-Distill-Qwen	BART	Total
ogbn-arxiv	91k(100%)	30k(33%)	45k(49%)	10k(11%)	91k(100%)	267k
ogbn-products	90k(46%)	81k(41%)	14k(7%)	24k(12%)	197k(100%)	406k

Impact of Profiling Models



Impact of Profiling Models



Training Parameters & Hardware Specifications

Parameter	Value
Learning Rate	3e-4
Batch Size	8
Gradient Accumulation Steps	2
Number of Epochs	1
Logging Steps	1
Optimizer	AdamW
Weight Decay	0.01
Warmup Steps	10
Random Seed	0
LoRA Rank	16
LoRA Alpha	16

Component	Specification
GPU	NVIDIA RTX A5000 (24GB memory)
Operating System	Ubuntu 20.04
Deep Learning Framework	PyTorch 2.5
CUDA Version	CUDA 12.2
Python Version	Python 3.11

Conclusion

Conclusion

- Propose an LLM-based novel framework by generating profiles to increase the size of fine-tuning data and carefully designing the fine-tuning data
- Compare variants of the fine-tuning data and each node's neighbor information thoroughly
  - The result reveals that using a small amount of neighboring node information for fine-tuning can help improve classification accuracy
- The experiment result shows that our method outperforms nine state-of-the-art method
- Our fine-tuned models are available on huggingface:
  - https://huggingface.co/xinyifang/ArxivLlama
  - https://huggingface.co/xinyifang/ProductsLlama

References

[1] Haixing Dai et al. “Auggpt: Leveraging chatgpt for text data augmentation”. In: IEEE Transactions on Big Data (2025)

[2] Arijit Ghosh Chowdhury and Aman Chadha. “Generative data augmentation using llms improves distributional robustness in question answering”. In: arXiv preprint arXiv:2309.06358 (2023)

[3] Vinay Samuel et al. “Can LLMs Augment Low-Resource Reading Comprehension Datasets? Opportunities and Challenges”. In: Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 4: Student Research Workshop)

[4] Varun Kumar, Ashutosh Choudhary, and Eunah Cho. “Data Augmentation using Pre-trained Transformer Models”. In: Proceedings of the 2nd Workshop on Life-long Learning for Spoken Language Systems. 2020, pp. 18–26

[5] Edward J Hu et al. “Lora: Low-rank adaptation of large language models.” In: ICLR 1.2 (2022), p. 3

[6] Tim Dettmers et al. “Qlora: Efficient finetuning of quantized llms”. In: Advances in neural information processing systems 36 (2023), pp. 10088–10115

[7] Damjan Kalajdzievski. A Rank Stabilization Scaling Factor for Fine-Tuning with LoRA. 2023. arXiv: 2312.03732 [cs.CL]

[8] Michael Han Daniel Han and Unsloth team. Unsloth. 2023. url: http://github.com/unslothai/unsloth

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Q&A

Thanks for listening!